Week 2 Lesson 2 — Analysing the Mental Health Dataset questions

30th October 2026

1 Preparing data for a machine learning model. — Additional questions

This notebook comes from the article Preparing data for a machine learning model.

Dataset in Kagle: Mental Health in Tech Survey

Article from: 30 May 2017. This is using a much older version of Pandas, so some of the code may not work with the current version. I will try to update it as I go along.

In the dataset there are 2 questions:

1. How does the frequency of mental health illness and attitudes towards mental health vary by geographic location?
2. What are the strongest predictors of mental health illness or certain attitudes towards mental health in the workplace?

1.1 Question 1

1.1.1 Frequency of mental health illness by geographic location

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("./data/week_2_survey.csv")
#num_columns = len(df.columns)
#pd.set_option("display.max_columns", num_columns)
df.head(30)
```

[1]:

	Timestamp	Age	Gender	Country	state	self_employed	\
0	2014-08-27 11:29:31	37	Female	United States	IL	NaN	
1	2014-08-27 11:29:37	44	M	United States	IN	NaN	
2	2014-08-27 11:29:44	32	Male	Canada	NaN	NaN	
3	2014-08-27 11:29:46	31	Male	United Kingdom	NaN	NaN	
4	2014-08-27 11:30:22	31	Male	United States	TX	NaN	
5	2014-08-27 11:31:22	33	Male	United States	TN	NaN	
6	2014-08-27 11:31:50	35	Female	United States	MI	NaN	
7	2014-08-27 11:32:05	39	M	Canada	NaN	NaN	
8	2014-08-27 11:32:39	42	Female	United States	IL	NaN	

```
9   2014-08-27 11:32:43   23    Male          Canada   NaN           NaN
10  2014-08-27 11:32:44   31    Male   United States    OH           NaN
11  2014-08-27 11:32:49   29    male        Bulgaria   NaN           NaN
12  2014-08-27 11:33:23   42  female   United States    CA           NaN
13  2014-08-27 11:33:26   36    Male   United States    CT           NaN
14  2014-08-27 11:33:57   27    Male          Canada   NaN           NaN
15  2014-08-27 11:34:00   29  female   United States    IL           NaN
16  2014-08-27 11:34:20   23    Male  United Kingdom   NaN           NaN
17  2014-08-27 11:34:37   32    Male   United States    TN           NaN
18  2014-08-27 11:34:53   46    male   United States    MD           Yes
19  2014-08-27 11:35:08   36    Male          France   NaN           Yes
20  2014-08-27 11:35:12   29    Male   United States    NY            No
21  2014-08-27 11:35:24   31    male   United States    NC           Yes
22  2014-08-27 11:35:48   46    Male   United States    MA            No
23  2014-08-27 11:36:24   41    Male   United States    IA            No
24  2014-08-27 11:36:48   33    male   United States    CA            No
25  2014-08-27 11:37:08   35    male   United States    TN            No
26  2014-08-27 11:37:23   33    male   United States    TN            No
27  2014-08-27 11:37:59   35  Female   United States    CA            No
28  2014-08-27 11:38:12   34    male   United States    OH            No
29  2014-08-27 11:38:18   37    Male  United Kingdom   NaN            No

   family_history treatment work_interfere    no_employees  …  \
0              No       Yes          Often            6-25  …
1              No        No         Rarely  More than 1000  …
2              No        No         Rarely            6-25  …
3             Yes       Yes          Often         26-100  …
4              No        No          Never        100-500  …
5             Yes        No      Sometimes            6-25  …
6             Yes       Yes      Sometimes             1-5  …
7              No        No          Never             1-5  …
8             Yes       Yes      Sometimes        100-500  …
9              No        No          Never         26-100  …
10             No       Yes      Sometimes            6-25  …
11             No        No          Never        100-500  …
12            Yes       Yes      Sometimes         26-100  …
13            Yes        No          Never       500-1000  …
14             No        No          Never            6-25  …
15            Yes       Yes         Rarely         26-100  …
16             No       Yes      Sometimes         26-100  …
17             No       Yes      Sometimes            6-25  …
18            Yes        No      Sometimes             1-5  …
19            Yes        No            NaN            6-25  …
20            Yes       Yes      Sometimes        100-500  …
21             No        No          Never             1-5  …
22             No       Yes          Often         26-100  …
23             No       Yes          Never  More than 1000  …
```

```
24            Yes       Yes         Rarely          26-100  …
25            Yes       Yes      Sometimes  More than 1000  …
26             No        No            NaN             1-5  …
27            Yes       Yes         Rarely            6-25  …
28             No       Yes      Sometimes          26-100  …
29             No        No      Sometimes            6-25  …

                leave mental_health_consequence phys_health_consequence  \
0        Somewhat easy                      No                       No
1           Don't know                   Maybe                       No
2   Somewhat difficult                      No                       No
3   Somewhat difficult                     Yes                      Yes
4           Don't know                      No                       No
5           Don't know                      No                       No
6   Somewhat difficult                   Maybe                    Maybe
7           Don't know                      No                       No
8        Very difficult                   Maybe                       No
9           Don't know                      No                       No
10          Don't know                      No                       No
11          Don't know                      No                       No
12  Somewhat difficult                     Yes                      Yes
13          Don't know                      No                       No
14       Somewhat easy                      No                       No
15       Somewhat easy                      No                       No
16           Very easy                   Maybe                       No
17          Don't know                   Maybe                       No
18           Very easy                      No                       No
19       Somewhat easy                      No                       No
20  Somewhat difficult                   Maybe                       No
21  Somewhat difficult                      No                       No
22          Don't know                   Maybe                       No
23          Don't know                   Maybe                       No
24          Don't know                      No                       No
25           Very easy                     Yes                       No
26          Don't know                   Maybe                    Maybe
27          Don't know                      No                       No
28  Somewhat difficult                      No                       No
29       Very difficult                     Yes                    Maybe

       coworkers    supervisor mental_health_interview phys_health_interview  \
0   Some of them           Yes                      No                  Maybe
1             No            No                      No                     No
2            Yes           Yes                     Yes                    Yes
3   Some of them            No                   Maybe                  Maybe
4   Some of them           Yes                     Yes                    Yes
5            Yes           Yes                      No                  Maybe
6   Some of them            No                      No                     No
```

```
7             No            No                     No                     No
8            Yes           Yes                     No                  Maybe
9            Yes           Yes                  Maybe                  Maybe
10  Some of them           Yes                     No                     No
11           Yes           Yes                    Yes                    Yes
12           Yes           Yes                  Maybe                  Maybe
13           Yes           Yes                     No                     No
14  Some of them  Some of them                  Maybe                    Yes
15           Yes  Some of them                  Maybe                  Maybe
16  Some of them            No                  Maybe                  Maybe
17  Some of them           Yes                     No                     No
18           Yes           Yes                     No                    Yes
19  Some of them  Some of them                  Maybe                  Maybe
20  Some of them  Some of them                     No                     No
21  Some of them  Some of them                     No                  Maybe
22  Some of them           Yes                     No                  Maybe
23            No            No                     No                    Yes
24           Yes           Yes                     No                    Yes
25  Some of them           Yes                     No                    Yes
26  Some of them            No                     No                     No
27           Yes           Yes                  Maybe                  Maybe
28  Some of them            No                     No                     No
29  Some of them            No                     No                  Maybe

   mental_vs_physical obs_consequence  \
0                 Yes              No
1         Don't know              No
2                  No              No
3                  No             Yes
4         Don't know              No
5         Don't know              No
6         Don't know              No
7                  No              No
8                  No              No
9                 Yes              No
10        Don't know              No
11        Don't know              No
12                 No             Yes
13        Don't know              No
14                Yes              No
15        Don't know              No
16                 No              No
17                 No              No
18                Yes             Yes
19        Don't know              No
20                 No              No
21                Yes              No
```

```
22                 No              No
23         Don't know              No
24         Don't know              No
25                 No              No
26         Don't know              No
27                Yes              No
28                 No              No
29                 No              No

                                                comments
0                                                    NaN
1                                                    NaN
2                                                    NaN
3                                                    NaN
4                                                    NaN
5                                                    NaN
6                                                    NaN
7                                                    NaN
8                                                    NaN
9                                                    NaN
10                                                   NaN
11                                                   NaN
12                                                   NaN
13  I'm not on my company's health insurance which…
14                                                   NaN
15  I have chronic low-level neurological issues t…
16  My company does provide healthcare but not to …
17                                                   NaN
18                                                   NaN
19                                                   NaN
20                                                   NaN
21                                                   NaN
22                                                   NaN
23                                                   NaN
24                Relatively new job. Ask again later
25  Sometimes I think  about using drugs for my me…
26                                                   NaN
27                                                   NaN
28                                                   NaN
29                                                   NaN

[30 rows x 27 columns]
```

You can't assume all participants have a mental health condition. This 2014 dataset also doesn't have a direct column confirming whether each respondent has one.

The closest field is `treatment`, which asks whether the respondent has sought treatment for a mental health condition. `work_interfere` asks how a condition interferes with work.

[2]:
```
columns_to_show = [
    "Country", "self_employed", "family_history", "work_interfere",
    "no_employees", "remote_work", "tech_company", "benefits",
    "care_options", "wellness_program", "seek_help", "anonymity",
    "leave", "mental_health_consequence", "phys_health_consequence",
    "coworkers", "supervisor", "mental_health_interview",
    "phys_health_interview", "mental_vs_physical", "obs_consequence",
]

df.loc[df["treatment"] == "Yes", columns_to_show]
```

[2]:

	Country	self_employed	family_history	work_interfere	\
0	United States	NaN	No	Often	
3	United Kingdom	NaN	Yes	Often	
6	United States	NaN	Yes	Sometimes	
8	United States	NaN	Yes	Sometimes	
10	United States	NaN	No	Sometimes	
…	…	…	…	…	
1252	United States	No	Yes	Sometimes	
1254	United Kingdom	No	No	NaN	
1255	United States	No	Yes	Often	
1256	United States	No	Yes	Sometimes	
1258	United States	No	Yes	Sometimes	

	no_employees	remote_work	tech_company	benefits	care_options	\
0	6-25	No	Yes	Yes	Not sure	
3	26-100	No	Yes	No	Yes	
6	1-5	Yes	Yes	No	No	
8	100-500	No	Yes	Yes	Yes	
10	6-25	Yes	Yes	Don't know	No	
…	…	…	…	…	…	
1252	100-500	Yes	Yes	Yes	Yes	
1254	26-100	No	Yes	No	No	
1255	26-100	Yes	Yes	Yes	Yes	
1256	More than 1000	No	Yes	Yes	Yes	
1258	26-100	No	No	Yes	Yes	

	wellness_program	…	anonymity	leave	\
0	No	…	Yes	Somewhat easy	
3	No	…	No	Somewhat difficult	
6	No	…	No	Somewhat difficult	
8	No	…	No	Very difficult	
10	No	…	Don't know	Don't know	
…	…	…	…	…	
1252	Yes	…	Yes	Don't know	
1254	No	…	Don't know	Somewhat easy	
1255	No	…	Yes	Somewhat difficult	

```
1256              No  …  Don't know  Somewhat difficult
1258              No  …         Yes          Don't know

     mental_health_consequence phys_health_consequence     coworkers  \
0                           No                       No  Some of them
3                          Yes                      Yes  Some of them
6                        Maybe                    Maybe  Some of them
8                        Maybe                       No           Yes
10                          No                       No  Some of them
…                           …                        …           …
1252                       Yes                       No  Some of them
1254                        No                       No  Some of them
1255                        No                       No  Some of them
1256                       Yes                      Yes            No
1258                     Maybe                       No  Some of them

        supervisor mental_health_interview phys_health_interview  \
0              Yes                      No                 Maybe
3               No                   Maybe                 Maybe
6               No                      No                    No
8              Yes                      No                 Maybe
10             Yes                      No                    No
…              …                        …                  …
1252            No                      No                 Maybe
1254  Some of them                      No                    No
1255           Yes                      No                    No
1256            No                      No                    No
1258            No                      No                    No

     mental_vs_physical obs_consequence
0                   Yes              No
3                    No             Yes
6            Don't know              No
8                    No              No
10           Don't know              No
…                   …              …
1252                 No              No
1254         Don't know              No
1255                Yes              No
1256                 No              No
1258         Don't know              No

[637 rows x 21 columns]
```

[3]:
```
country_counts = (
    df.loc[df["treatment"] == "Yes"]
    .groupby("Country")
```

```
    .size()
    .reset_index(name="Sought treatment")
    .sort_values("Sought treatment", ascending=False)
)

country_counts
```

[3]:

	Country	Sought treatment
26	United States	410
25	United Kingdom	93
5	Canada	37
10	Germany	21
12	Ireland	13
0	Australia	13
17	Netherlands	9
18	New Zealand	5
22	South Africa	4
19	Poland	4
11	India	3
24	Switzerland	3
6	Croatia	2
7	Denmark	2
4	Bulgaria	2
9	France	2
3	Brazil	2
23	Sweden	2
20	Singapore	1
21	Slovenia	1
14	Japan	1
16	Moldova	1
15	Mexico	1
1	Bahamas, The	1
13	Italy	1
8	Finland	1
2	Belgium	1
27	Zimbabwe	1

[4]:
```
country_summary = (
    df.assign(sought_treatment=df["treatment"].map({"Yes": 1, "No": 0}))
    .groupby("Country")["sought_treatment"]
    .agg(
        Respondents="count",
        Sought_treatment="sum",
        Percentage="mean",
    )
)
```

```
country_summary["Percentage"] = (
    country_summary["Percentage"] * 100
).round(1)

country_summary.sort_values("Percentage", ascending=False)
```

[4]:

Country	Respondents	Sought_treatment	Percentage
Japan	1	1	100.0
Croatia	2	2	100.0
Slovenia	1	1	100.0
Moldova	1	1	100.0
Denmark	2	2	100.0
Zimbabwe	1	1	100.0
Bahamas, The	1	1	100.0
South Africa	6	4	66.7
New Zealand	8	5	62.5
Australia	21	13	61.9
Poland	7	4	57.1
United States	751	410	54.6
Canada	72	37	51.4
United Kingdom	185	93	50.3
Bulgaria	4	2	50.0
Ireland	27	13	48.1
Germany	45	21	46.7
Switzerland	7	3	42.9
Mexico	3	1	33.3
Brazil	6	2	33.3
Netherlands	27	9	33.3
Finland	3	1	33.3
India	10	3	30.0
Sweden	7	2	28.6
Singapore	4	1	25.0
Belgium	6	1	16.7
France	13	2	15.4
Italy	7	1	14.3
Israel	5	0	0.0
Uruguay	1	0	0.0
China	1	0	0.0
Colombia	2	0	0.0
Thailand	1	0	0.0
Greece	2	0	0.0
Hungary	1	0	0.0
Spain	1	0	0.0
Costa Rica	1	0	0.0
Latvia	1	0	0.0
Austria	3	0	0.0

Russia	3	0	0.0
Romania	1	0	0.0
Portugal	2	0	0.0
Czech Republic	1	0	0.0
Philippines	1	0	0.0
Georgia	1	0	0.0
Nigeria	1	0	0.0
Bosnia and Herzegovina	1	0	0.0
Norway	1	0	0.0

[5]:
```
min_respondents = 20

grouped_summary = (
    country_summary.assign(
        Country_group=country_summary.index.where(
            country_summary["Respondents"] >= min_respondents,
            "Other",
        )
    )
    .groupby("Country_group")[["Respondents", "Sought_treatment"]]
    .sum()
    .rename_axis("Country")
)

grouped_summary["Percentage"] = (
    grouped_summary["Sought_treatment"]
    / grouped_summary["Respondents"]
    * 100
).round(1)

grouped_summary.sort_values("Respondents", ascending=False)
```

[5]:

Country	Respondents	Sought_treatment	Percentage
United States	751	410	54.6
United Kingdom	185	93	50.3
Other	131	41	31.3
Canada	72	37	51.4
Germany	45	21	46.7
Ireland	27	13	48.1
Netherlands	27	9	33.3
Australia	21	13	61.9

Again: **`treatment == "No"` does not mean "no mental illness."**

This is because the survey question only asks whether the respondent has sought treatment for a mental health condition, not whether they have a mental health condition. Therefore, a respondent who answers "No" to the treatment question may still have a mental health condition that they

have not sought treatment for.

1.1.2 Attitudes towards mental health in the workplace

We compare attitudes towards mental health across countries using the following survey columns:

Column	What it measures
`mental_health_consequence`	Whether respondents expect negative consequences from discussing mental health with their employer
`coworkers`	Willingness to discuss mental health with coworkers
`supervisor`	Willingness to discuss mental health with a direct supervisor
`mental_vs_physical`	Whether respondents believe their employer takes mental health as seriously as physical health

This analysis includes **all respondents**, regardless of whether they have sought treatment.

Countries with fewer than **20 respondents** are combined into **Other**. For each question, we calculate the percentage of each response within each country group, excluding missing answers. Percentages therefore total approximately 100% per group.

A stacked horizontal bar chart shows how response distributions differ geographically. Percentages for **Other** are calculated from the pooled respondents, rather than by averaging country percentages.

Responses must be interpreted according to the question. For example, **Yes** to `mental_health_consequence` indicates an expectation of negative consequences, whereas **Yes** to `supervisor` indicates willingness to discuss mental health with a supervisor.

These results describe reported attitudes and workplace perceptions among survey respondents. They do not establish the attitudes of entire countries or the prevalence of mental illness.

[6]:
```
attitude_meanings = {
    "mental_health_consequence": (
        "Expected negative consequences of discussing mental health with an␣
 ↪employer"
    ),
    "coworkers": "Willingness to discuss mental health with coworkers",
    "supervisor": "Willingness to discuss mental health with your direct␣
 ↪supervisor(s)",
    "mental_vs_physical": (
        "Whether the employer takes mental health as seriously as physical␣
 ↪health"
    ),
}
```

```
def table_attitute_percentages(attitude, min_respondents=20):
    assert isinstance(attitude, str), "attitude must be a string"
    assert attitude in attitude_meanings.keys(), "attitude must be one of␣
 ↪'mental_health_consequence', 'coworkers', 'supervisor', or 'seek_help'"

    #attitude = "mental_health_consequence"

    # Use all respondents to examine attitudes.
    country_counts = df["Country"].value_counts()

    country_groups = df["Country"].where(
        df["Country"].map(country_counts) >= min_respondents,
        "Other",
    )

    # Within each country group, calculate the response percentages.
    attitude_percentages = (
        pd.crosstab(
            country_groups,
            df[attitude],
            normalize="index",
        )
        .mul(100)
        .round(1)
    )

    return attitude_percentages

def display_stacked_graph(attitude_percentages_df, attitude):
    assert isinstance(attitude, str), "attitude must be a string"

    attitude_percentages_df.plot.barh(
        stacked=True,
        figsize=(10, 6),
    )

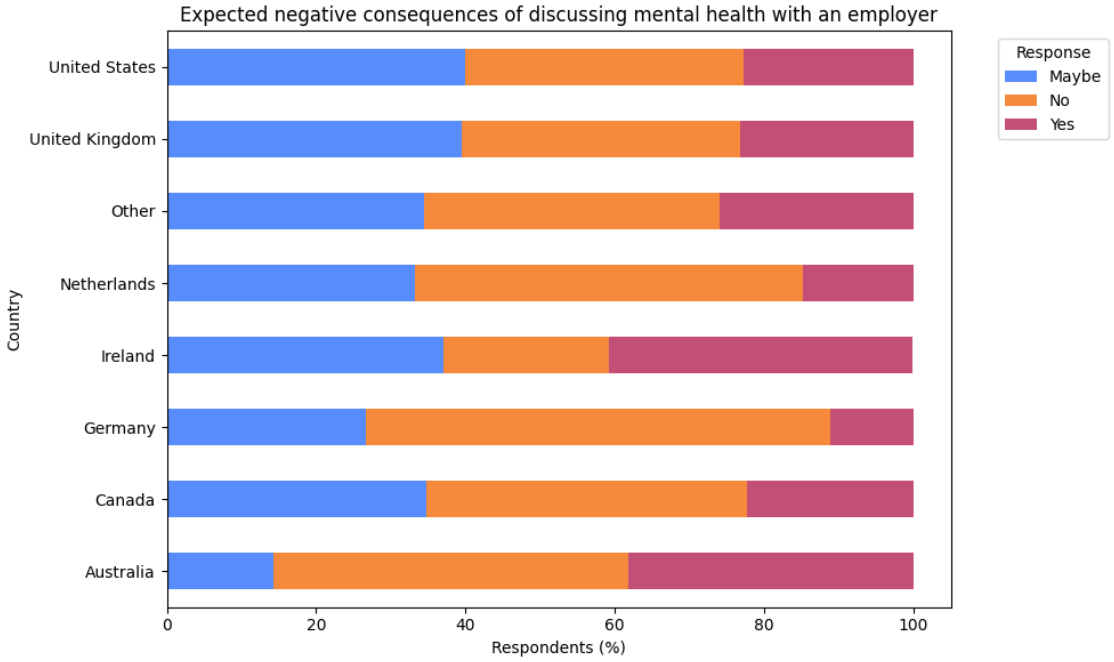
    plt.xlabel("Respondents (%)")
    plt.ylabel("Country")
    plt.title(attitude_meanings[attitude])
    plt.legend(title="Response", bbox_to_anchor=(1.05, 1), loc="upper left")
    plt.tight_layout()
    plt.show()

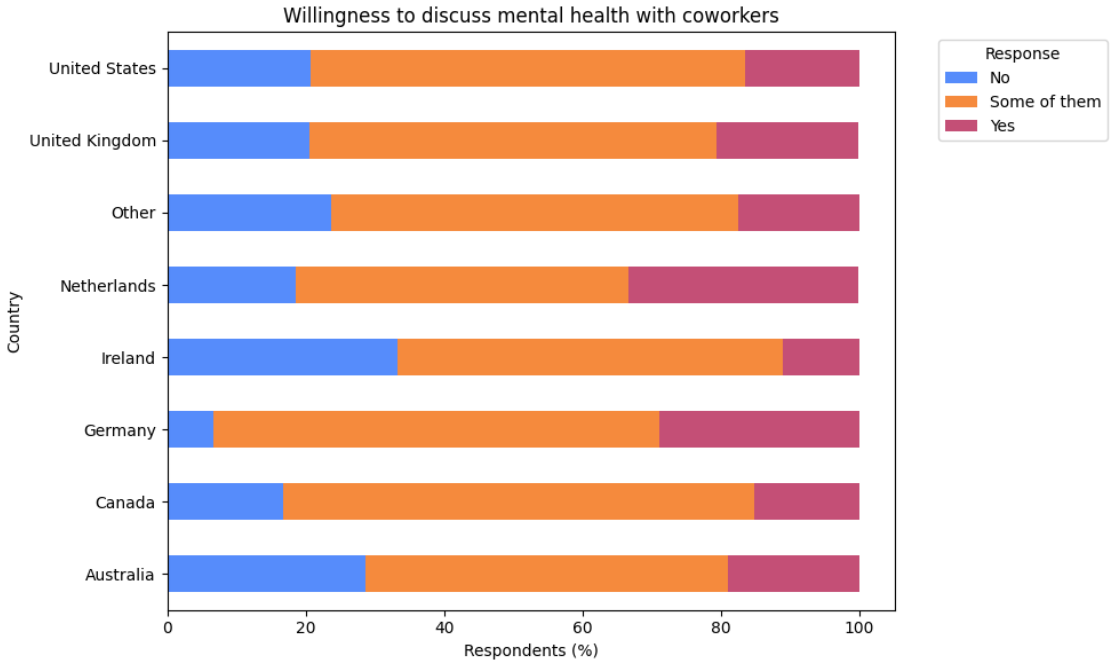
attitude_percentages = table_attitute_percentages("mental_health_consequence")
display(attitude_percentages)
```

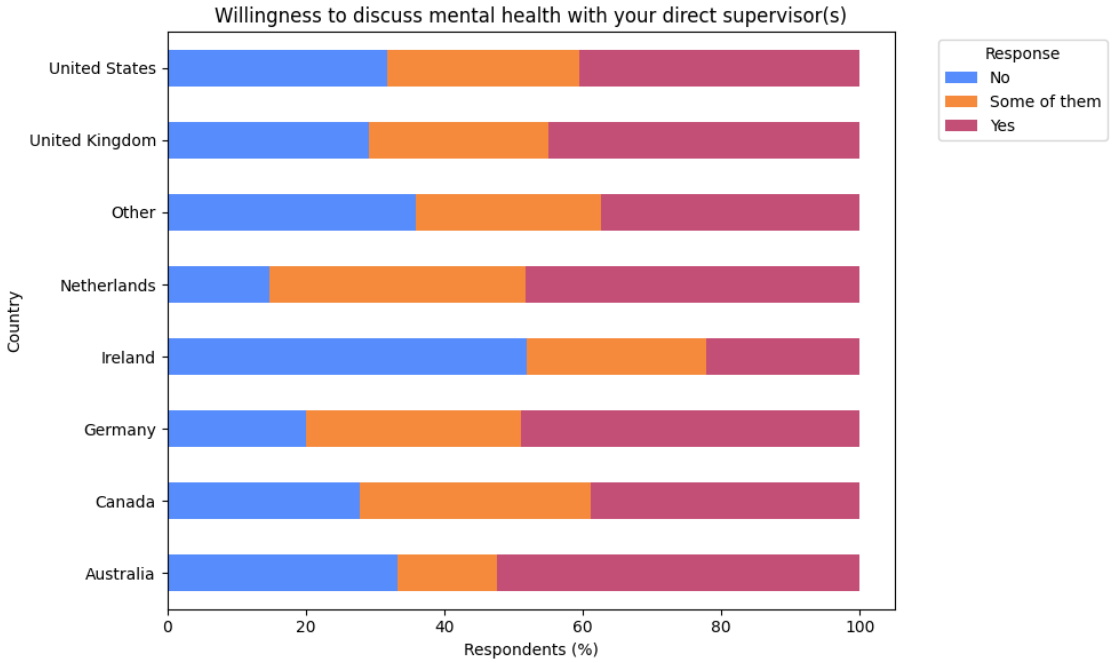
```
mental_health_consequence  Maybe    No   Yes
Country
```

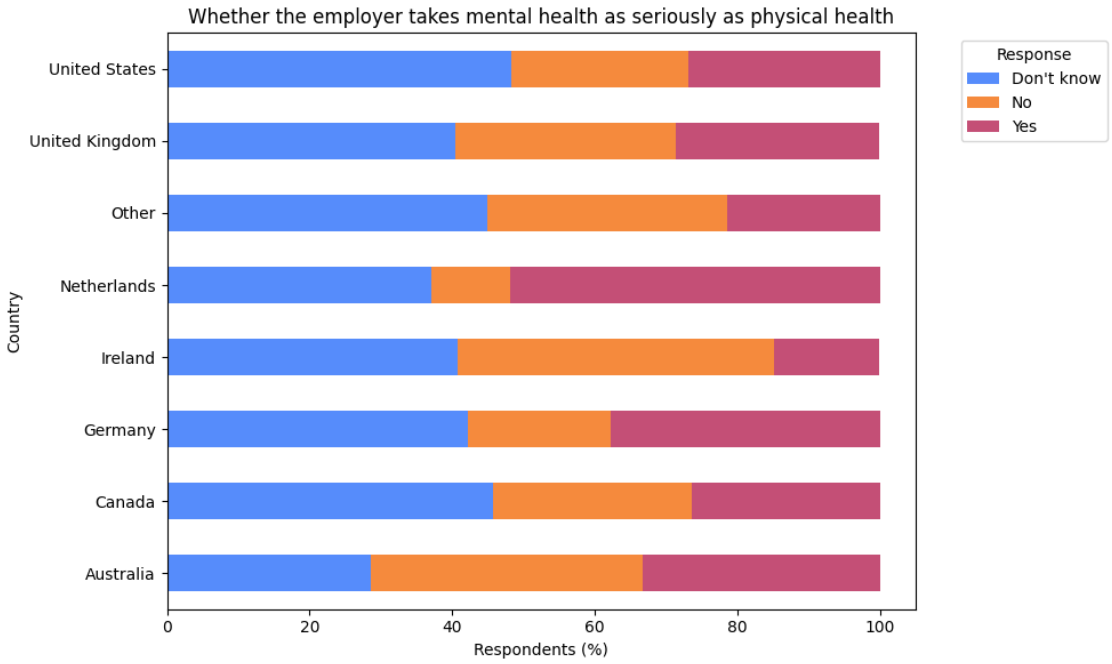
```
Australia                14.3  47.6  38.1
Canada                   34.7  43.1  22.2
Germany                  26.7  62.2  11.1
Ireland                  37.0  22.2  40.7
Netherlands              33.3  51.9  14.8
Other                    34.4  39.7  26.0
United Kingdom           39.5  37.3  23.2
United States            39.9  37.3  22.8
```

[7]:
```
for attitude in attitude_meanings.keys():
    attitude_df = table_attitute_percentages(attitude)
    display_stacked_graph(attitude_df, attitude)
```




Willingness to discuss mental health with coworkers
United States
United Kingdom
Other
Netherlands
Ireland
Germany
Canada
Australia
Country
0
20
40
60
80
100
Respondents (%)
Response
No
Some of them
Yes


Willingness to discuss mental health with your direct supervisor(s)
United States
United Kingdom
Other
Netherlands
Ireland
Germany
Canada
Australia
Country
0
20
40
60
80
100
Respondents (%)
Response
No
Some of them
Yes



1.2 Question 2

What are the strongest predictors of mental health illness or certain attitudes towards mental health in the workplace?

1.2.1 Predictors of treatment seeking and workplace attitudes

This analysis examines which demographic, family-history, and workplace characteristics help predict whether respondents sought mental health treatment or report particular workplace attitudes.

The `treatment` column measures treatment seeking, not the presence of a diagnosed mental illness.

We will evaluate predictive models on held-out data and rank predictors using permutation importance. Variables whose shuffling produces the largest decrease in predictive performance will be considered the most important for that model.

These findings describe predictive associations among survey respondents. They do not demonstrate that a characteristic causes mental illness, treatment seeking, or a particular attitude.

1.2.2 Example code

This trains a **logistic regression model** to predict treatment seeking, compares it with a baseline, and ranks predictors. It assumes your survey is loaded as df.

Preprocessing is fitted only on the training data to avoid leakage.

Encoding company size The `no_employees` column contains ordered categories:

`1-5` → `6-25` → `26-100` → `100-500` → `500-1000` → `More than 1000`

Although ordinal encoding preserves this order, it introduces an additional assumption when used in logistic regression:

Encoding	Assumption in logistic regression
One-hot — the current approach	Each size category can have its own relationship with treatment seeking
Ordinal	Every step up a category has the same effect on the predicted log-odds

Encoding the categories as integers from 0 to 5 means that moving from **1–5 to 6–25** has the same modeled effect as moving from **500–1000 to more than 1,000**, despite the unequal company-size ranges. It also imposes a consistently increasing or decreasing relationship, unless the coefficient is zero.

One-hot encoding accommodates patterns such as respondents in medium-sized companies being more likely to seek treatment than those in either smaller or larger companies.

For this analysis, we retain **one-hot encoding as the baseline** because there are only six categories and we have not established a consistently increasing or decreasing relationship.

Ordinal encoding could be evaluated as an alternative using cross-validation on the training data. The low permutation importance of `no_employees` alone does not demonstrate that its encoding should change.

[8]:
```
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.dummy import DummyClassifier
from sklearn.metrics import (
    accuracy_score,
    balanced_accuracy_score,
    roc_auc_score,
    classification_report,
    ConfusionMatrixDisplay,
)
from sklearn.inspection import permutation_importance

# 1. Select predictors and the outcome.
numeric_columns = ["Age"]
categorical_columns = [
    "Gender",
    "Country",
    "family_history",
    "self_employed",
    "no_employees",
```

```
    "remote_work",
    "tech_company",
    "benefits",
    "care_options",
    "wellness_program",
    "seek_help",
    "anonymity",
    "leave",
]

# Keep only respondents with a valid target response.
model_data = df.loc[df["treatment"].isin(["Yes", "No"])].copy()

X = model_data[numeric_columns + categorical_columns].copy()
y = model_data["treatment"].map({"No": 0, "Yes": 1})

# Treat ages outside this analysis's adult range as missing.
# The training-data median will fill them later.
X["Age"] = pd.to_numeric(X["Age"], errors="raise")
# Filter between 18 and 80 years old.
X.loc[~X["Age"].between(18, 80), "Age"] = np.nan

# 2. Reserve 25% for evaluation.
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42,
    stratify=y, # If not None, data is split in a stratified fashion, using␣
 ↪this as the class labels.
)

# 3. Define preprocessing.
numeric_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler()),
])

categorical_pipeline = Pipeline([
    ("imputer", SimpleImputer(
        strategy="constant",
        fill_value="Missing",
    )),
    ("encoder", OneHotEncoder(handle_unknown="ignore")),
])

preprocessor = ColumnTransformer([
```

```
    ("numeric", numeric_pipeline, numeric_columns),
    ("categorical", categorical_pipeline, categorical_columns),
])

# 4. Train the complete pipeline.
model = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", LogisticRegression(max_iter=2000)),
])

model.fit(X_train, y_train)
```

[8]:
```
Pipeline(steps=[('preprocessor',
                 ColumnTransformer(transformers=[('numeric',
                                                  Pipeline(steps=[('imputer',
SimpleImputer(strategy='median')),
                                                                  ('scaler',
StandardScaler())]),
                                                  ['Age']),
                                                 ('categorical',
                                                  Pipeline(steps=[('imputer',
SimpleImputer(fill_value='Missing',
 strategy='constant')),
                                                                  ('encoder',
OneHotEncoder(handle_unknown='ignore'))]),
                                                  ['Gender', 'Country',
                                                   'family_history',
                                                   'self_employed',
                                                   'no_employees',
                                                   'remote_work',
                                                   'tech_company', 'benefits',
                                                   'care_options',
                                                   'wellness_program',
                                                   'seek_help', 'anonymity',
                                                   'leave'])])),
                ('classifier', LogisticRegression(max_iter=2000))])
```

Finally, estimate which original columns contribute most to the model's test performance:

[9]:
```
importance = permutation_importance(
    model,
    X_test,
    y_test,
    scoring="roc_auc",
    n_repeats=20,
    random_state=42,
    n_jobs=1,
)
```

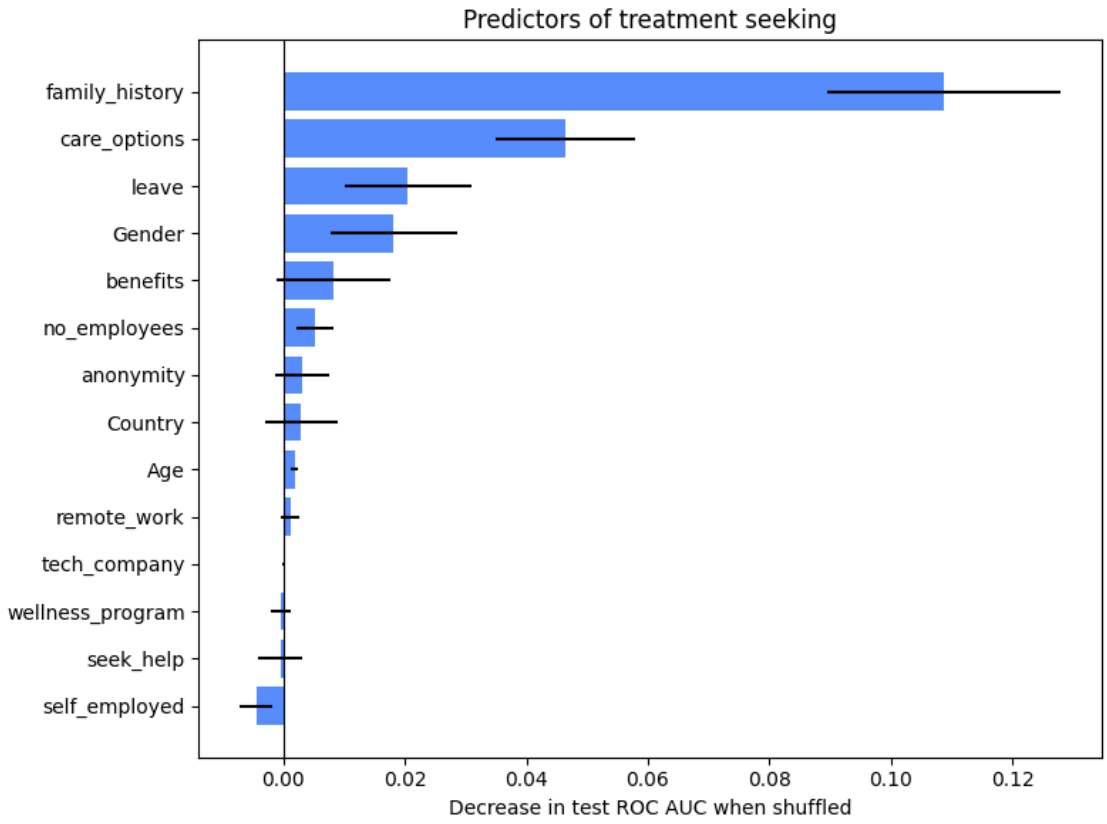
```
predictor_importance = pd.DataFrame({
    "Predictor": X_test.columns,
    "Mean AUC decrease": importance.importances_mean,
    "Standard deviation": importance.importances_std,
}).sort_values("Mean AUC decrease", ascending=False)

display(predictor_importance.round(4))

plot_data = predictor_importance.sort_values("Mean AUC decrease")

fig, ax = plt.subplots(figsize=(8, 6))
ax.barh(
    plot_data["Predictor"],
    plot_data["Mean AUC decrease"],
    xerr=plot_data["Standard deviation"],
)
ax.axvline(0, color="black", linewidth=0.8)
ax.set(
    title="Predictors of treatment seeking",
    xlabel="Decrease in test ROC AUC when shuffled",
)
plt.tight_layout()
plt.show()
```

	Predictor	Mean AUC decrease	Standard deviation
3	family_history	0.1088	0.0192
9	care_options	0.0465	0.0115
13	leave	0.0205	0.0104
1	Gender	0.0181	0.0105
8	benefits	0.0083	0.0094
5	no_employees	0.0052	0.0030
12	anonymity	0.0031	0.0044
2	Country	0.0030	0.0060
0	Age	0.0018	0.0005
6	remote_work	0.0011	0.0016
7	tech_company	-0.0000	0.0002
10	wellness_program	-0.0004	0.0016
11	seek_help	-0.0005	0.0037
4	self_employed	-0.0045	0.0027



1.2.3 Understanding the predictor importance chart

This chart shows how much the trained model relies on each column to predict whether a respondent sought mental health treatment.

We measure this using **permutation importance**:

1. Evaluate the model on the original test data.
2. Randomly rearrange—or **shuffle**—the values in one column among respondents. Keep the other columns and actual treatment responses unchanged.
3. Evaluate the same model again, without retraining it.
4. Measure how much its performance decreased.

Shuffling preserves the values in a column but disrupts their connection to the correct respondents. If predictions become worse, this suggests the model was using that column's information.

Performance is measured using **ROC AUC**, which describes how well the model ranks respondents who sought treatment above those who did not. A score of 0.5 represents chance-level ranking, while 1.0 represents perfect ranking.

1.2.4 Reading the blue bars

Each blue bar shows the **average decrease in test ROC AUC** across 20 shuffles of that column.

Bar position	Meaning
Longer bar to the right of zero	Shuffling substantially reduced performance, suggesting the model relies more on this column
Short bar near zero	Shuffling made little difference, suggesting little additional predictive contribution in this model
Bar to the left of zero	Shuffling improved performance slightly, suggesting the column may add noise or that the result reflects variation in this test sample

For example, if ROC AUC falls from **0.80 to 0.70** after shuffling a column, its importance for that shuffle is **0.10**. This is a change in ROC AUC, not a 10% decrease in accuracy.

A positive importance does **not** mean that a feature increases treatment seeking. It means that the feature helps the model make predictions. The chart does not show which answers make treatment seeking more or less likely.

1.2.5 Reading the black error bars

The black horizontal lines show **one standard deviation above and below the average importance** across the 20 shuffles.

- **Shorter lines** mean the measured importance was similar across shuffles.
- **Longer lines** mean the measured importance varied more.
- **Lines extending across zero** suggest the measured contribution is small relative to its variation.

These lines describe variation caused by shuffling the same test data. They are **not confidence intervals** and do not capture all uncertainty, such as what would happen with a different sample of respondents.

1.2.6 Why a low importance does not mean "irrelevant"

Some predictors contain overlapping information. For example, workplace benefits and awareness of care options may be related.

If one column is shuffled, the model may still obtain similar information from another column. Its performance may therefore change little, even though the shuffled column contains useful information.

Importance depends on the model, the other predictors included, and the data used for evaluation.

1.2.7 What we can conclude

These results describe predictive associations with **treatment seeking**, not the presence or causes of mental illness. Someone may have a mental health condition without having sought treatment.

The results also come from a single division of the data into training and test sets. A different split could produce different scores or predictor rankings.

If we use these findings to choose columns or adjust the model, this test set becomes part of the model-development process. Final performance should then be evaluated on a separate, untouched

test set to check whether the improvements also work on respondents whose data did not influence those decisions.